EXTENDING AN N-TRUNCATED SIMPLICIAL OBJECT BY A REPRESENTABLE FUNCTOR AND THE COSKELETON ADJUNCTION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the adjunction bijection is constructed in both directions (the representability step and the restriction step) and the author leaves only the mutual-inverse verification to the reader ('We leave it to the reader to show these are mutually inverse constructions', source line 2697) - accepted per the precedent of cards L46149 and L46175 (both constructions explicit, inverse check omitted); no other verification is omitted.

Editorial note. The label lemma-work-out names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The truncation limit identifications. Same chapter, \label{lemma-trivial-cosk} and \label{lemma-formula-limit}. Proof lines 2668-2678: U_i = lim over ([Delta]/[i])_{<= n}^opp for i <= n, and U_{n+1} = lim over ([Delta]/[n+1])_{<= n}^opp.
- The coskeleton construction template. Same chapter, \label{lemma-existence-cosk}. Proof lines 2678-2686: the maps U-tilde(phi) are defined exactly as in this lemma, giving an (n+1)-truncated simplicial object with sk_n U-tilde = U.
- Representability of the compatibility relation. Statement lines 2644-2662. U_{n+1} represents the functor of families (f_0, ..., f_{n+1}) in Mor(T, U_n) satisfying d^n_{j-1} f_i = d^n_i f_j for all i < j.

Source reference. Original reference 0183: sources/stacks-simplicial/pinned-original-source.tex, source lines 2451–2451. Prerequisite pointer invoked at proof line(s) 2681; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0186: sources/stacks-simplicial/pinned-original-source.tex, source lines 2554–2554. Prerequisite pointer invoked at proof line(s) 2673; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0184: sources/stacks-simplicial/pinned-original-source.tex, source lines 2509–2509. Prerequisite pointer invoked at proof line(s) 2669; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let n be an integer ≥ 1. Let U be a n-truncated simplicial object of $\mathcal{C}$. Consider the contravariant functor from $\mathcal{C}$ to Sets which associates to an object T the set*

$$\{(f_0, \ldots, f_{n+1}) \in \mathrm{Mor}_{\mathcal{C}}(T, U_n) \mid d^n_{j-1} \circ f_i = d^n_i \circ f_j \ \forall\ 0 \leq i < j \leq n+1\}$$

If this functor is representable by some object U_{n+1} of $\mathcal{C}$, then there exists an $(n+1)$-truncated simplicial object $\tilde{U}$, with $sk_n\tilde{U} = U$ and $\tilde{U}_{n+1} = U_{n+1}$ such that the following adjointness holds

$$\mathrm{Mor}_{Simp_{n+1}(\mathcal{C})}(V, \tilde{U}) = \mathrm{Mor}_{Simp_n(\mathcal{C})}(sk_n V, U)$$

Proof. By Lemma 0184 there are identifications

$$U_i = \lim_{(\Delta/[i])^{opp}_{\leq n}} U(i)$$

for $0 \leq i \leq n$. By Lemma 0186 we have

$$U_{n+1} = \lim_{(\Delta/[n+1])^{opp}_{\leq n}} U(n).$$

Thus we may define for any $\varphi : [i] \to [j]$ with $i, j \leq n+1$ the corresponding map $\tilde{U}(\varphi) : \tilde{U}_j \to \tilde{U}_i$ exactly as in Lemma 0183. This defines an $(n+1)$-truncated simplicial object $\tilde{U}$ with $\mathrm{sk}_n\tilde{U} = U$.

To see the adjointness we argue as follows. Given any element $\gamma : \mathrm{sk}_n V \to U$ of the right hand side of the formula consider the morphisms $f_i = \gamma_n \circ d^{n+1}_i : V_{n+1} \to V_n \to U_n$. These clearly satisfy the relations $d^n_{j-1} \circ f_i = d^n_i \circ f_j$ and hence define a unique morphism $V_{n+1} \to U_{n+1}$ by our choice of U_{n+1}. Conversely, given a morphism $\gamma' : V \to \tilde{U}$ of the left hand side we can simply restrict to $\Delta_{\leq n}$ to get an element of the right hand side. We leave it to the reader to show these are mutually inverse constructions. $\square$